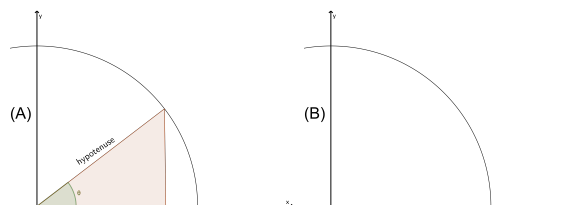


Problem 3

Problem 3

Problem 1 A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see figure (A)). One of its sides lies on the x -axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle θ with the x -axis.



Make sure to label the picture.

- Draw 2 more examples of such a triangle in the figure (B).
- Express the area of such a triangle as a function of θ and state its domain.
- Find the value of θ which maximizes the area in part (b). Show your work and justify your answer. [2]